

Where Qwen’s layer-0 “–” lean comes from

Supplementary analysis record — sign-flip circuit paper. Qwen2.5-72B-Instruct, 4×4 determinant, held-out set (147 errors + 141 corrects); Gemma-3-27B comparison in §5. Sept 2, 2026.

1. The observation

Logit-lens curves at the probe position (the token right before the model writes the sign of interest) show a lead of about **+4.5 logits for “–” over “+” at layer 0** on 98% of problems — regardless of which sign is correct and regardless of which sign the model later writes (+4.50 mean when it wrote “–”, –4.4 when it wrote “+”: the same “–” bias either way). This layer-0 bump also fools the peak-depth metric on 33 problems in Fig. 1 (shown there as open circles). This report explains where the bump comes from.

2. How the measurement works

The model has two token tables: the **embedding table** (each token → a 8,192-number vector, used at the input) and the **unembedding table** (each token → a vector the final state is compared against, via dot product, to score the next token). We take the fixed direction $\mathbf{u} = \text{unembed}(\text{“–”}) - \text{unembed}(\text{“+”})$ (token ids 481/488) and dot it with a vector $\mathbf{x}$; a positive value means $\mathbf{x}$ pushes “–” more than “+”. At layer 0 the residual at the probe position is nothing but the embedding row of the token sitting there, so the layer-0 lean of a problem is (up to the final RMSNorm, which we apply) the table lookup $\text{lean}(\mathbf{t}) = \mathbf{u} \cdot \text{embed}(\mathbf{t})$ for its probe-position token $\mathbf{t}$.

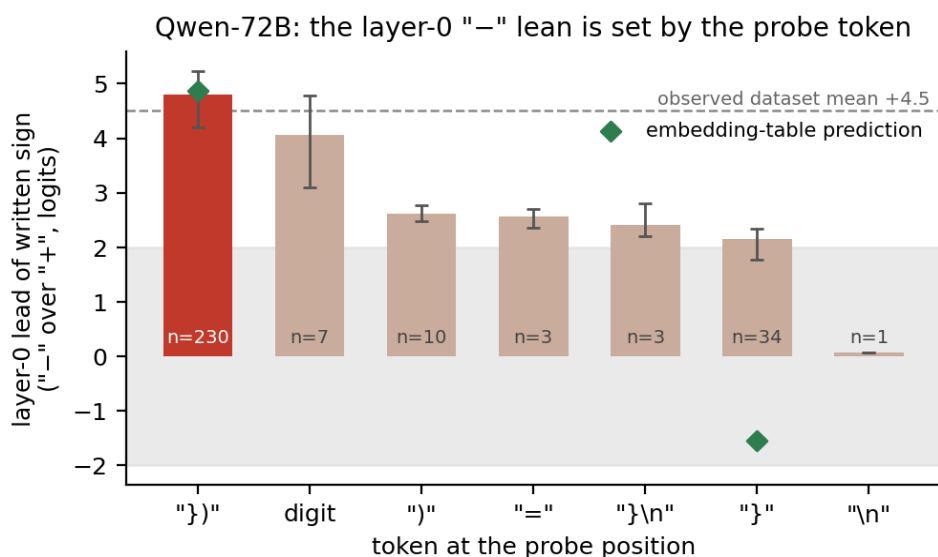
3. Three tests, run in order

Test 1 — global weight geometry: ruled out. (`qwen_embed_geometry.py`) Across 24,998 sampled vocabulary tokens, only 49.6% (raw) / 49.3% (RMSNorm) lean “–” — chance. The lean is **not** a blanket property of the weights.

Test 2 — text composition: contributes, but falls short. (`qwen_text_lean.py`) Tokenizing all 288 question texts (cut at the sign position) gives 108,229 token occurrences; **68.2%** of that mass leans “–” (baseline 49.3%), weighted mean **+0.99**. Drivers: digit “1” (+6.05, $n=4,232$), literal “–” tokens (+3.43, $n=6,073$), LaTeX markup. But $+0.99 < +4.5$: the average text cannot explain the observed magnitude.

Test 3 — the probe-position token: closes the gap. We matched each of the 288 records to its tokenized text and read off the token at the probe position. The per-problem layer-0 lean is nearly a lookup table of that one token:

token at probe	n	mean L0 lead	range	embedding-table prediction
"}"	230 (80%)	+4.80	+4.21 to +5.22	+4.87
digits (1,2,3,8,9)	7	+3.10 to +4.68		+6.05 ('1'), +2.15 ('2'), +0.78 ('3')
")"	10	+2.61	+2.48 to +2.78	not computed
" ="	3	+2.56	+2.35 to +2.70	not computed
"}n"	3	+2.41	+2.20 to +2.80	not computed
"}"	34	+2.15	+1.77 to +2.33	–1.54
newline	1	+0.07		not computed



4×4 determinant, held-out (n=288). Bars: mean layer-0 lead grouped by the token at the probe position; whiskers: min-max. Green diamonds: lean predicted from the embedding table alone (known for "}") +4.87 and "}" -1.54). 80% of probes land on "})" — its prediction (+4.87) matches the observed +4.80 almost exactly, explaining the dataset mean of +4.5. Gray band: |lead| < 2 (noise). Dashed line the dataset mean +4.5.

Figure: mean layer-0 lead of the written sign (sign-flipped so positive = leans “-”), grouped by the token at the probe position; whiskers show min-max, green diamonds the pure embedding-table prediction, gray band |lead| < 2 (noise), dashed line the dataset mean +4.5.

4. Reading of the result

The +4.5 lean is not a property of the problem — it is a property of one token. 80% of probes land on “}”, the LaTeX close of terms like $\det(A_{11})$ that sits right before the sign. Its embedding row alone predicts +4.87; the observed group mean is +4.80, with a tight range (4.21–5.22) across 230 problems, errors and corrects, both written signs. In plain terms: in Qwen’s training data, after “})” in math text a “-” simply followed more often than a “+”, and this bigram statistic is baked into how the two token tables align.

A small context effect exists, but is secondary. The “}” group is the check: its embedding predicts -1.54 (leans “+”), yet the 34 problems show +2.15 — so layer-0 computation (block-0 attention over the minus-heavy context) adds roughly +3.7 there. Shallow copying is real but does not drive the headline number. The single problem probed at a newline shows no lean (+0.07), as predicted.

Cross-domain replication (integration by parts). If the lean values are input-independent weight constants, the same probe token must show the same lean in a different problem domain. The 76 held-out IBP problems provide the test: their sign never follows “})” — it follows “)” (n=68) or “)” (n=8) — and the per-token values replicate within ~0.2 logits of the determinant runs:

probe token	det 4×4 (mean, range)	IBP (mean, range)
"})"	+2.61 [+2.48, +2.78]	+2.86 [+2.40, +3.21]
"})"	+2.15 [+1.77, +2.33]	+1.97 [+1.77, +2.18]

All 76/76 IBP problems lean “-” at layer 0, but at only +2.77 on average — because “)” and “)” carry smaller lookup entries than “})” (+4.87). This also explains why the IBP appendix figure shows **zero layer-0 artifacts**: a +2.8 bump is too weak to outrank any real late peak, so the peak-depth metric is never fooled there, unlike the 33 determinant cases. Same baked-in lean, cheaper token, hidden symptom.

Cross-domain replication (arithmetic). The 48-problem shared arithmetic control set extends the test to a third domain and a second model. Qwen leans “-” on **48/48** problems (mean +2.89, all >2), again set by probe-token constants: “[\n” n=31 at +2.66, “[\n” at +2.86, “boxed” at +4.42, “=” at +3.18. Llama-70B on the

identical problems shows **no clear lean at all** (0/48 above +2, mean +0.19) — matching its faint determinant profile (8% clear). The domain gradient — $\text{det} +4.5 > \text{arith} +2.9 \approx \text{IBP} +2.8$ — exists only because 80% of determinant probes land on the expensive token “}””; arithmetic and IBP probes land on cheaper tokens. Only the determinant lean is big enough to fool the peak-depth metric, which is why layer-0 artifacts appear only in the determinant panel of Fig. 1.

Implication for Fig. 1. This confirms treating the 33 layer-0 peaks as an embedding artifact, not an early decision: their curves go negative through the mid layers and only cross back positive at 91–96% depth, so the true decision is still late. It also refines the paper’s claim: the lean is not a global weight property; it tracks the token composition of math text, and its magnitude at the probe position is set almost entirely by the identity of the probe token itself.

5. Gemma comparison: same symptom, opposite mechanism

Gemma-3-27B shows an early “-” lean too (layers 8–13, clear on 95% of problems, up to +68 logits, erased by ~L15). Running the same per-problem analysis on its 37 determinant records (gemma_context_lean.py, fully offline) gives the opposite diagnosis:

No layer-0 lean. L0 is positive on only 14/37 (~chance) — the embedding-lookup mechanism that explains Qwen does not apply. **The L8–13 lean is universal (37/37) and graded by context:**

text right before the probe	n	mean peak lean (L8-13)	mean L0
literal “-” within 5 chars	17	+16.6 (up to +68.1)	+6.3
other mid-computation	6	+11.0	+0.8
right after computed number	2	+9.3	+6.2
after “}”	5	+5.7	-1.7
after “)” / cofactor C	3	+3.1	-2.9
after “)” (env close)	4	+2.1	-3.2

The lean correlates with the count of literal “-” characters in the last 5 chars before the probe ($r = +0.42$). Those minus characters are the model’s own expansion text: of 137 in the last 50 chars across all records, 102 are negative numbers (matrix entries, operands like $(-4) * (-900)$), 21 subtraction operators, 11 negative intermediate results ($= -2765$). **Verdict: copying, not lookup** — Gemma’s early layers shallow-copy nearby “-” tokens to pre-load the sign guess, then late layers overwrite it. Qwen pre-loads “-” from a weight-table bigram (after “}”); Gemma from copying its own recent minus ink — two roads to the same bad prior. Caveat: $n=37$, small categories; indicative ordering, not tight.

6. Reproducibility

Scripts and result JSONs: paper/appendix/lean_tests/ in the paper repository (qwen_embed_geometry.py, qwen_text_lean.py, qwen_probe_token_lean.py, gemma_context_lean.py, all *_results.json, qwen_tokenizer.json; the last two run fully offline). Logit-lens records: Qwen/heldout_4x4_DET/s01_logit_lens/det_4x4_{error,correct}_expA_logit_lens.json. Question texts: Qwen/raw_data_4x4_DET/heldout/*.csv. IBP cross-check: Qwen/heldout_IBP/s01_logit_lens/ + Qwen/raw_data_IBP/heldout/*.csv. Gemma data: paper/appendix/lean_tests/gemma_data/ (expA_logit_lens.json + gemma_determinant_experiment_ready.xlsx) — the Gemma analysis in §5 was run on this earlier 37-problem determinant subset (a separate, workshop-phase run), not the main discovery/held-out Gemma sets elsewhere in this repository. Weight rows fetched from HuggingFace by HTTP Range request (bf16, model Qwen/Qwen2.5-72B-Instruct, untied embeddings); sign tokens 481 (“-”) / 488 (“+”); RMSNorm with the model’s final-norm gain applied before the dot product. Arithmetic (48 shared problems): {Llama,Qwen}/arith_control/expA_logit_lens.json.